

Industrial Organization - Homework II

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Due: Start of class November 6

You can work in groups of up to 2 people.

Background

In this homework assignment, you will be coding up a replication of Rust (1987). Specifically, I want you to attempt to replicate (or, at least, produce estimates that closely resemble) the estimates for ‘Group 4’ buses in Table IX of the paper (on page 1021) for the discount factor $\beta = 0.9999$ and fixed point dimension = 90. The estimates I want you to reproduce are for the replacement cost RC and the cost function θ_{11} ; more details are below.

Descriptions of the Variables

- **Bus_ID:** ID of the bus
- **period:** period since the start of the data
- **state:** binned state of mileage (i.e., in 5000 mile increments)
- **mileage:** raw mileage number
- **decision:** Harold Zurcher’s decision to replace the bus engine. If he does replace the engine, this =1 and you can see that the mileage will reset to 0 the following period.

1. Rust (1987) replication

Consider the Rust (1987) model that we discussed in detail in class (see the lecture slides if you need to be reminded of all the details). Assume the following specifications, which correspond to the model estimated in Table IX of the paper:

- Cost function $c(x, \theta_1) = 0.001\theta_{11}x$
- Discount parameter: $\beta = 0.9999$

- Idiosyncratic errors are i.i.d. logit
- Fixed point state space discretization is dimension 90
- Transition parameters given as follows:
 - $\theta_{30} = 0.3919$
 - $\theta_{31} = 0.5953$
- Parameters that are unknown and need to be estimated from the data:
 - RC: replacement cost
 - θ_{11} : parameter in the cost function (given above)

Instructions

- (a) Estimate the replacement cost RC and the cost function parameter θ_{11} by maximum likelihood using the nested fixed-point method for computing the value functions. (More specifically, follow the Rust (1987) algorithm to replicate/find close numbers to the estimates for Group 4 buses in Table IX.) No need to compute standard errors.

Attached to this homework, I have included the code used to replicate (estimate) the replacement cost RC and the cost function parameter θ_{11}

Our estimated results are as follows:

$$\begin{aligned}\theta_{11_R} &= 2.2931(\text{true} : 2.2930) \\ RC_R &= 10.0773(\text{true} : 10.0750)\end{aligned}$$

- (b) Provide pseudo-code for an alternative CCP method to the nested fixed-point method. Discuss one benefit and one limitation of this CCP method as compared to the full solution method you used in (a.). (To emphasize: no need to write the exact code or compute values with this method, pseudo-code is enough.)

An alternative method could be the one discussed in Hotz & Miller (1993)¹. This algorithm consists of the following four steps:

1. Estimate the Conditional Choice Probabilities (CCPs) directly from the data.

¹Hotz, V. Joseph, and Robert A. Miller. "Conditional choice probabilities and the estimation of dynamic models." The Review of Economic Studies 60.3 (1993): 497-529.

2. Apply the Hotz & Miller inversion.
Using the estimated CCPs, compute the expected value function $\hat{V}$ via the following inversion formula:

$$\hat{V} = \left[I - \beta \sum_i P_i * T_i \right]^{-1} * \left(\sum_i P_i * [u_i + \mathbb{E}[\epsilon_i]] \right)$$

Here, I is the identity matrix, β is the discount factor, P_i are the probabilities of choosing action i , T_i is the transition matrix, u_i is the utility for action i , and $\mathbb{E}[\epsilon_i]$ represents the expected value of the error term.

3. Compute the Expected Probability (EP) from the Expected Value (EV). Calculate the probability of taking action $a = 1$ using the following formula:

$$\hat{\text{Pr}}(i = 1; \theta) = \frac{\exp(u_1 + \beta T_1 \hat{V})}{\sum_i \exp(u_i + \beta T_i \hat{V})}$$

4. Compute (and maximize) the objective function (log-likelihood).
The likelihood function $\mathcal{L}(\theta)$ is computed as:

$$\mathcal{L}(\theta) = \prod_{t=1}^T \left(\hat{\text{Pr}}(a = 1 \mid x_t; \theta) \mathbb{1}(a_t = 1) \cdot \left(1 - \hat{\text{Pr}}(a = 1 \mid x_t; \theta) \right) \mathbb{1}(a_t = 0) \right)$$

In this approach, we estimate the CCPs directly from the data. However, this cannot be done using the fixed point dimension of 90. Instead, the dimensions need to be adjusted to match those of the data. In this case, the data dimension comprises 78 intervals.

Attached to this homework is the code we used to implement this algorithm to estimate the replacement cost RC and the cost function parameter θ_{11} . Our results are as follows:

$$\begin{aligned} \theta_{11_{HM1}} &= 10.8636(\text{true} : 2.2930) \\ RC_{HM1} &= 27.1784(\text{true} : 10.0750) \end{aligned}$$

The results are significantly off, which is expected given the limited number of observations in many of these intervals.

To address this issue, one approach is to reduce the number of intervals to ensure each interval contains sufficient data. Although this choice might seem somewhat arbitrary, and changing the bin size renders the results not directly comparable, we conducted an experiment by increasing the bin size from 5000 to 50000 miles. After applying this adjustment, the updated results are as follows:

$$\begin{aligned} \theta_{11_{HM2}} &= 76.6003 \\ RC_{HM2} &= 6.8812 \end{aligned}$$

The results change significantly and remain far from the true values. Although they are not directly comparable due to the change in bin size, this suggests that the method, aside from being somewhat arbitrary, may not effectively address the underlying issue.

Another possible approach is to assume a specific functional form for the CCPs, such as a linear structure, and then estimate the empirical CCPs under this assumption.

In this approach, we performed a logit regression of the decision variable with respect to the state to generate predicted probabilities of action for each of the 78 intervals. These predicted probabilities were then used to construct the CCPs. After applying this adjustment, the updated results are as follows:

$$\begin{aligned}\theta_{11_{HM3}} &= 10.9452(\text{true} : 2.2930) \\ RC_{HM3} &= 27.1105(\text{true} : 10.0750)\end{aligned}$$

The results are still significantly off and, interestingly, similar to those obtained when calculating the CCPs directly for each interval from the data (HM1). These estimation issues can be attributed to the fact that the Hotz & Miller method is typically recommended for very large samples, a condition not met by the data provided for this homework.